Reiwa 6 (2024)

Date submitted: 18/11/2024

Fundamental Labs I Report

(School of Science and Engineering, College of Engineering Systems)

☐ 7 System control engineering basic labs
☑ 8 Basics of linear systems using operational amplifier
☐ 9 Diodes and transistors
☐ 10 Basics of logic circuits and computers
☐ 11 DC motor manufacturing and control
☐ 12 Mechanisms and mechanical elements

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(3) Nikos Tan (4)_____
(5)_____ (6)_____

	1st week	2nd week	Report submission
Lab date	Month/Day	Month/Day	Received stamp
Lab participant's stamp with seal Non-participants leave blank	工学システム学類 基礎実験 22-1s 線形システム	工学システム学類 基礎実験 22-1s 線形システム	
Latecomer arrival time			

Use this sheet as the report cover page.
Reports are submitted by the date/time and to the location designated for each theme.
Submission deadline: Month: 11 Date: 24 (), Time: 11 : 40 Submission location: Manaba
NOTE: This should be collected and saved by each teaching assistant (TA) after the mandatory information has been initially filled in during the first lab. After their participation in the second lab has been understood, this sheet is returned to students who are instructed to use it as the cover page of their report.

Chapter 8 Report

Chapter 8 Week 1 Report

Purpose of the lab:

An operational amplifier is an important active circuit element that is used for analog amplification and analog operation in electric circuits. In this lab, in order to understand the operating characteristics of operational amplifier circuits, we learn about analog basic amplifier circuits and analog operation circuits , such as those used for addition, subtraction, and calculus..

In this week's lab, we learned about inverting amplifier circuits and the following circuits based on the amplifier and inverting amplifier circuits in order : integrator circuit, summing circuit, subtraction circuit, and determined an unknown constructed circuit as a derivative or differentiator circuit.

Assignment 1 (Inverting Amplifier Circuit)

Experiment details

For this assignment, our team used an electronic breadboard, 2 9V batteries as a ±9V power supply, 10 k Ω and 100 k Ω resistor to construct an inverting amplifier circuit. We then inserted a DC power supply for input voltage V_{in} and incremented it from -1 to 1 V.

The circuit is constructed as follows:

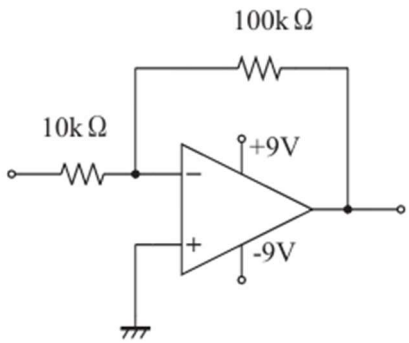


Figure 1 Inverting amplifier circuit

With R1=10 k Ω and Rf = 100 k Ω

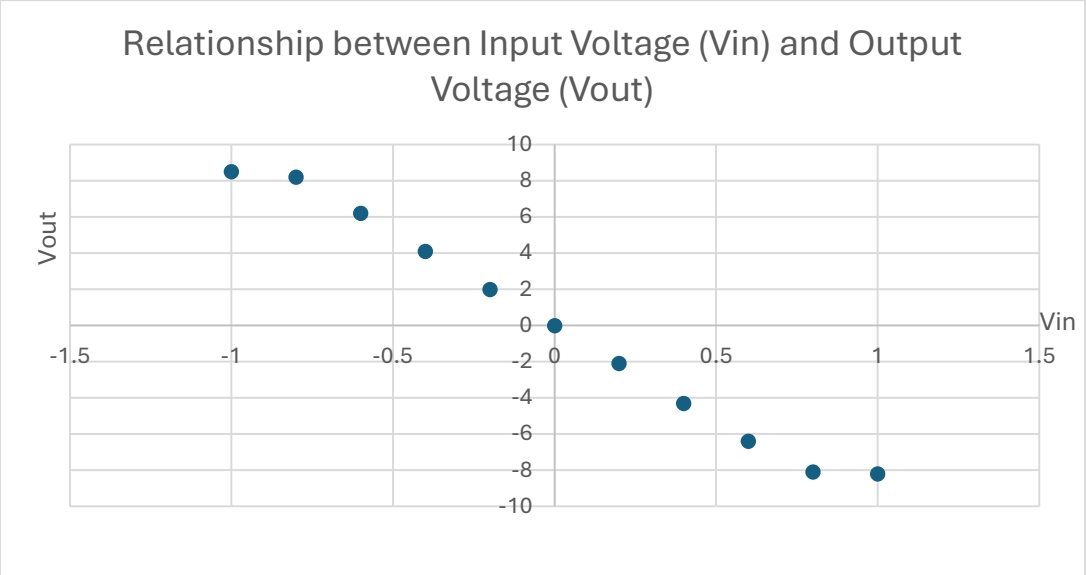
Result

The input voltages(V_{in}) and output voltages (V_{out}) are as recorded in the following table:

Vin	Vout

-1	8.5
-0.8	8.2
-0.6	6.2
-0.4	4.1
-0.2	2
0	0
0.2	-2.1
0.4	-4.3
0.6	-6.4
0.8	-8.1
1	-8.2
Table 1	

And as the following graph constructed in Excel, which shows an almost linear relationship between Vin and Vout except for the converging points



Graph 1

Discussion and analysis

Calculating the gain

$$G = \quad v_2 / v_1 \quad = \quad -\frac{I_1 R_f}{I_1 R_1} = \quad -R_f / R_1$$

which should be roughly G= -10, the voltage theoretical output should be $v_2 \quad = -10v_1$

However, the graph converges near Vin = -1 V and Vin = 1 V

From the oral exam, there are 3 characteristics of the graph:

1. Vout = a negative multiple of Vin (which is the definition of an inverting amplifier circuit)
2. The graph converges as mentioned above. The reason for this is that the amplifier is weak when the output voltage value is near that of the battery power supplies (±9V)
3. If Δ Vin = 0.2 V, Δ Vout = -2V is not levelling or converging near ±9V

Assignment 2 (Integrator Amplifier Circuit)

Experiment details

We constructed the integrator amplifier circuit according to the following electric circuit diagram:

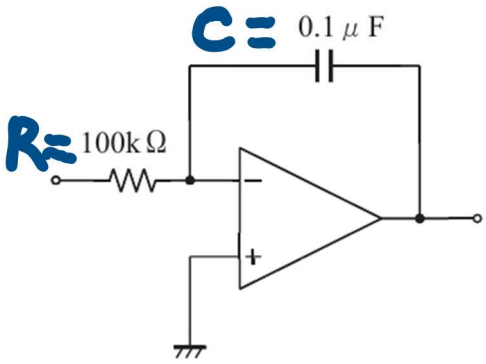


Figure 2 Integrator Amplifier circuit diagram

We used a 0.1 microfarad capacitor, a 100 k Ω resistor, and inserted an AC power supply for the input voltage Vin. The integrator amplifier is connected to 9V battery source and ground. We then inputted a square wave with a frequency of approximately 200Hz. We then measured and recorded the waveforms of both the input signal Vin and output signal Vout using an oscilloscope.

Result

The output is integrated (hence the integrator circuit name). The input is a square wave, so the output is a triangular wave. The output is a negative integral of the input. Hence, if we have a constant positive voltage on the input, then the output will linearly decrease.

The graph of Vin and Vout recorded in the oscilloscope is identical to Graph 2 drawn on the graph paper.

Discussion and analysis

The voltage output of an integrator circuit is

$$V_{out} = -\frac{1}{R_1 C}\int_0^t V_{in}\,dt = -\int_0^t V_{in}\frac{dt}{R_1 C}$$

From the equation above, the gain of the circuit is

$$G = -\frac{1}{CR_1} = \frac{1}{100X \quad {}^{3}(0.1\times10^{-6})} = -1000$$

Hence, the output is the negative integral of the circuit. Because the input is a square wave, the output is a triangular wave.

Assignment 3 (Summing amplifier circuit)

Experiment details

We constructed an electrical circuit according to the following circuit diagram:

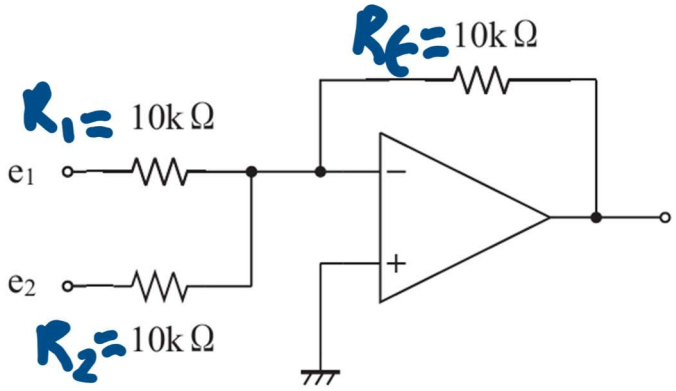


Figure 3 Summing circuit diagram

For the input e1, we used a 0.5V DC signal, and for the input e2, we used a sine wave with a frequency of approximately 1kHz. The summing amplifier is connected to 9V battery source and ground.

Results and discussion

The output is e1+ e2, and then inverted like the circuits in assignments 1 and 2. The output inverted upside down. The graph on the oscilloscope is identical to the graph 3 drawn on the graph paper. Thus, the DC signal is then added to the AC input to produce larger output voltage, and then inverted to produce lesser voltage.

The gain of the summing circuit is:

For e1: $G = -\frac{R_f}{R_1} = -\frac{10\text{k}\,\Omega}{10\text{k}\,\Omega} = 1$

For e2: $G = -\frac{R_f}{R_2} = -\frac{10\ \ \Omega}{10\text{k}\,\Omega} = 1$

Assignment 4(Subtraction amplifier circuit)

We constructed the subtraction circuit according to the following diagram:

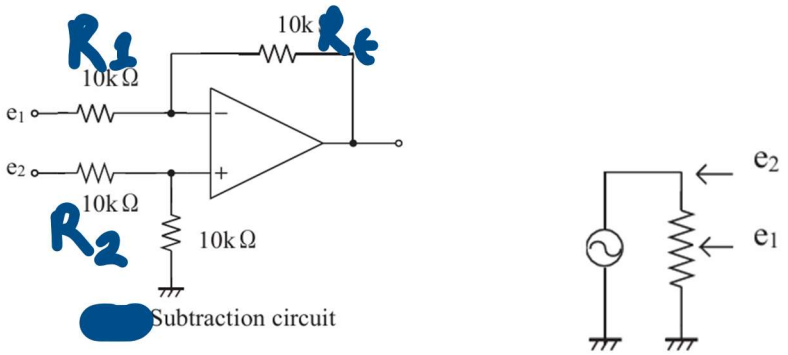


Figure 4

The subtraction circuit is connected to 9V battery source and ground.

For the experiment, we used a sine wave with a frequency of approximately 1kHz as the input signal. We split the input into two parts: reducing the voltage of one part using a variable resistor to create e1, and using the other part without reducing the voltage as e2. We then measured and recorded the waveforms of the input signals se1 and se2 using an oscilloscope.

Result and Discussion

The circuit is an inverting amplifier because the output is Vout = e2-e1 = – (e1+e2). Hence, this is still an inverting circuit. The graph of the oscilloscope is identical to the graph drawn on the graph paper.

The gain of the subtraction amplifier circuit is

$G = \frac{R_f}{R_{12}} = \frac{10\text{k}\,\Omega}{10\text{k}\,\Omega} = 1$

$$v_{out} = \frac{R_f}{R_1}(e_2 - e_1) = e_2 - e_1$$

Assignment 5 (Derivative or differentiator circuit)

Experiment details

We constructed the circuit according to the following diagram. Our team individually chose the value of the resistor and the capacitor. For the circuit, we used a 10 kΩ resisor and a 1.4 microfarad capacitor.

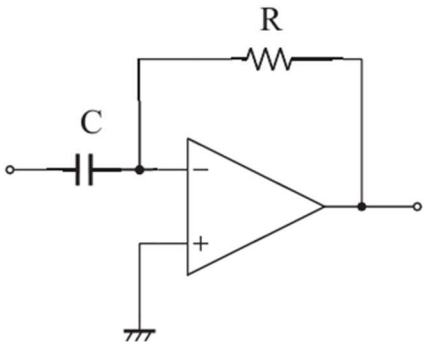


Figure 5 Differentiator amplifier circuit diagram

From the experiment, we had to determine the name of the circuit, and observe the output of the circuit for input waves sine wave and triangular wave using an oscillpscope.

Result and Discussion

From the output, we determined that it is a differentiator or derivative circuit. From the circuit diagram, it can be seen that the derivative circuit is the inverse of the integrator circuit because in the integrator circuit diagram in Figure 2, the resistor is connected to the input of the battery and the capacitor is connected to the output of the battery, while in the derivative circuit diagram in Figure 5, the capacitor is connected to the input of the battery while the resistor is connected to the output of the battery.

For the input triangular wave, the output on the diagram is triangular with noise in the graph (Graph 5), especially on the peak.

There is noise on the peak because the points in the peak is non-differentiable (cannot be differentiated)

For the input sine wave, the amplitude of the output is larger than the amplitude of the input in the graph of the differentiator circuit (Graph 5). This is because with the input $Vin = \sin(\omega t)$,

$$Vout = \frac{dVin}{dt} = \omega cos\omega t$$

The gain of the circuit is $G = -R_f\ C = -(10 \times 10^3)(1.4 \times 10^{-6}) = 0.014$

Vout = $-R_f\ C \frac{dVin}{dt}$

Week 2

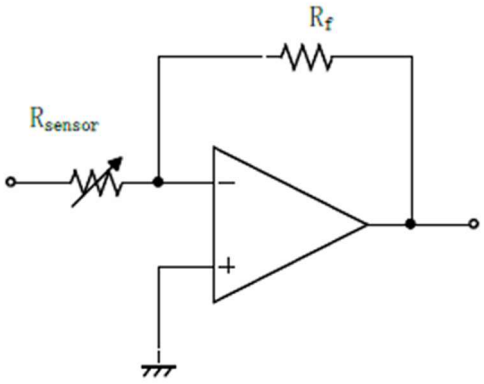
Purpose of the lab:

To measure physical phenomena, analog-digital conversion between sensors and information devices is essential. In this chapter, we focus on the basic principles and characteristics of contact position measurement using a contact position sensor, an amplifier circuit using pressure-voltage conversion and an operational amplifier, and the principle and application of analog-digital conversion.

Assignment 1

In Assignment 1, a SoftPot contact position sensor 100 mm was used to detect the position pressed by a finger from the voltage (variable resistance in Figure 6 corresponds to a sensor), and then we configured the circuit. We then used a DC voltage input of -5V.

The circuit is constructed according to the following diagram, connecting the sensor to an inverting amplifier circuit. For the inverting amplifier circuit, we connected it to 9V battery power supply like in week 1 assignment 1.



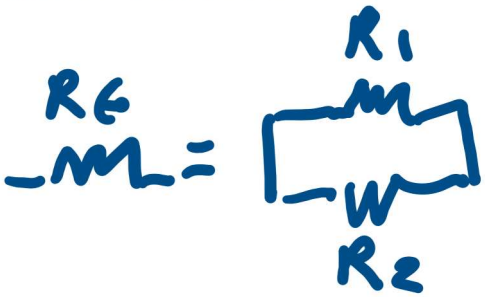


Figure 6

With Rsensor = the resistance provided by the sensor

We then measured the output using an oscilloscope, and adjusted the value of Rf so that the value had to be within ±5V. Thus, we made the value of Rf small by constructing parallel resistors 33 Ω and 55 Ω

Rf =1/ ((1/ 33) +(1/55)) = 20.65 Ω

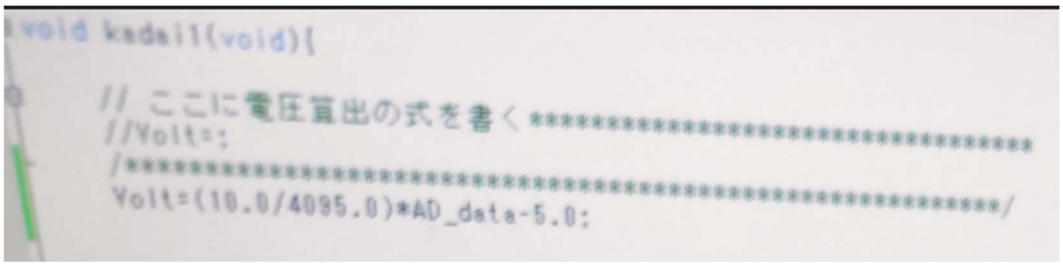
Assignment 2

Experiment details

The above circuit was used to determine the distance-voltage characteristics according to the AD-conversion board. We connected the output of the circuit to the AD conversion board, and then modified the C++ program function 'kadai1()' to convert the AD-converted digital values (global variable variable 'AD_data') into voltages. We then run the modified program and make contact with the contact position sensor at distances from 0 to 10cm in 0.5 cm increments.

The AD conversion board has resolution 12 bit, so the resolution is 10 / $(2^{12} - 1)$ = 10/ 4095

The following program function code is used for assignment 2:




```
void kadai1(void){
//volt=:
volt= (10:4095.0)*AD_data-5.0;
}
```

The entire code for Week 2 assignments 2,3,and 4 for AD and DA conversion will be included at the appendix.

Results and discussion

The distances from 0 to 10cm in 0.5 increments we made on the contact sensor and the corresponding output voltage are recorded in the table below using Excel:

Table 2 Distances we made on the contact sensor and the corresponding output

Distance	Volt
0	0.1013
0.5	0.1038
1	0.1087
1.5	0.1136
2	0.1233
2.5	0.1306
3	0.138
3.5	0.1477
4	0.16
4.5	0.1746
5	0.1893
5.5	0.2112
6	0.2405
6.5	0.265
7	0.2991
7.5	0.3578
8	0.4408
8.5	0.58
9	0.84
9.5	1.16
10	3.2979

Using Excel, we then created a graph and applied the polynomial trendline with the following equations with volt at the horizontal axis and distance at the vertical axis:

We split the graph into two parts, the graph for voltage when volt <0.58 V and when volt >0.58V separately, for more accuracy:

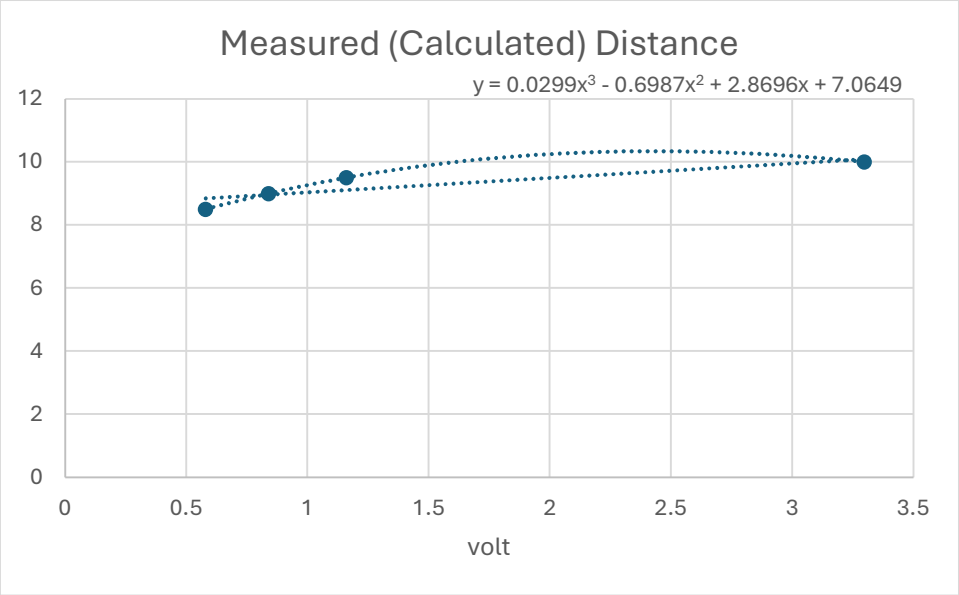
.

Graph 6 The relationship between distance made on the contact sensor and volt output in the AD conversion board when the volt is smaller than 0.58 V

When volt (x) <0.58, the equation for distance(y) as a function of volt is $y = -914.21x^4 + 1402.9x^3 - 793.61x^2 + 204.87x - 13.639$

This is a quartic graph which suggests the relationship between volt on the ADC converter and the distance on the contact sensor is nonlinear.

While when volt >0.58 V, the graph and the equation are the following:



Graph 7 The relationship between distance made on the contact sensor and volt output in the AD conversion board when the volt is larger than 0.58 V

For volt(x)>0.58 V, the equation for distance(y)as a function of volt is y = $0.0299x^3 - 0.6987x^2 + 2.8696x + 7.0649$

This is a cubic equation, so again the relationship between the distance and voltage is nonlinear.

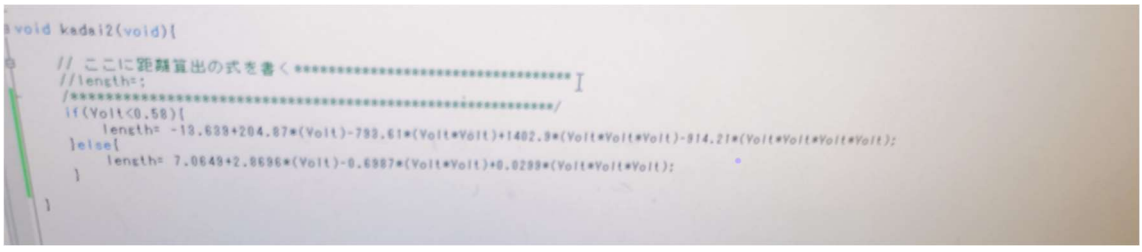
Assignment 3

Purpose To make a digital meter recording the distance we made contact with the contact sensor

Experiment details

We rewrote part of the program, program function ‘kadai(2)’ using the global variable ‘Volt’, which stores the voltage values, to write the distance calculation formula derived from Assignment 2. We rewrote the program to construct a digital distance meter.

The program function ‘kadai(2)’ is below:



```
void kadai2(void){

//length=:

if(Volt<0.58){

    length = -13.639+ 204.87 *(Volt) -793.61*(Volt*Volt)+1402.9*(Volt*Volt*Volt)-
914.21(Volt*Volt*Volt*Volt);

}

else{

length = 7.0649 + 2.8696*(Volt) -0.6987*(Volt*Volt) +0.0299*(Volt*Volt*Volt);

}
```

We then ran the program, made contact with the contact sensor at distances from 0 to 10cm in 1 cm increments, and read and recorded the displayed distances. We then compared the digital meter against the analog distance recorded by a ruler in 1cm increments.

Result

The real distance and the measured(calculated) distance is recorded in the graph below, Graph 8 using Excel. We used linear approximation or linear trendline to generate the graph. The values are almost the same with very small error with R^2 value of 0.9987 (near 1)

Error still exists due to 3 reasons:

1) Modelling error: No 100% model can be created
2) Human error: Error due to human contact with the sensor and mechanical error in the sensor due to temperature and outside influence.
3) Conversion value from Analog to Digital error: such as sampling error, quantization error, mistakes in the gain, and mistakes in the sampling rate

Some ADC errors and effect of errors on ADC performance: [2]

ADCs encounter various origins of inaccuracies and errors, this include offset errors, gain errors, linearity errors, quantization errors, noise, and jitter. Each of these factors exerts distinctive influences on the performance of the ADC:

Offset Errors: Offset error manifests as a consistent disparity between the genuine output of the ADC and the theoretical output. Consequently, the entire transfer function experiences a displacement. In scenarios necessitating precise direct current (DC)

precision, like the acquisition of sensor data, offset errors possess the potential to yield substantial deviations from the authentic values.

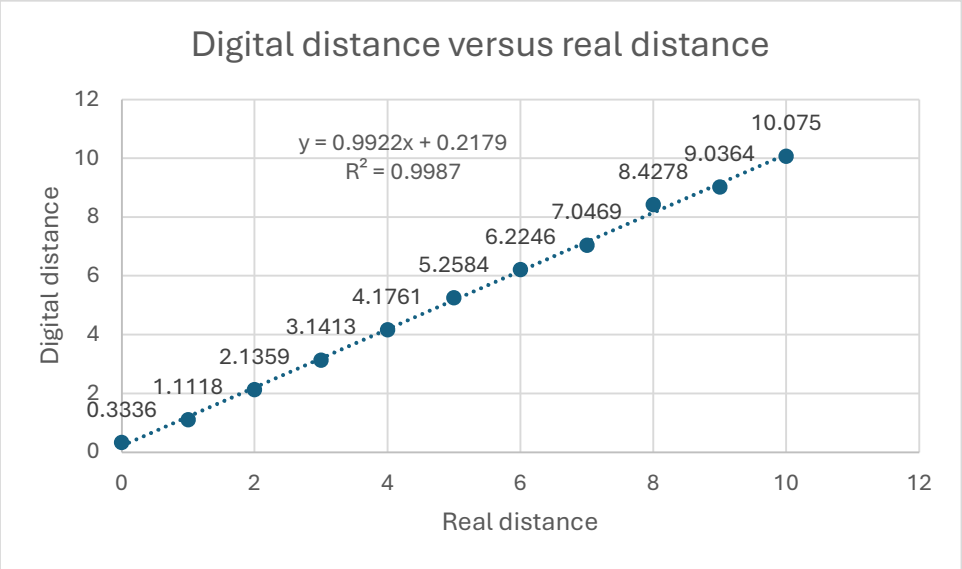
Gain Errors: Gain error corresponds to a divergence in the gradient of the ADC's transfer function from the ideal gradient. This discrepancy might lead to an amplified or attenuated representation of the input signal. The significance of gain errors is more pronounced in applications such as spectroscopy, wherein meticulous amplitude information holds crucial importance.

Linearity Errors: Errors related to linearity, encompassing Differential Non-Linearity (DNL) and Integral Non-Linearity (INL), relate to deviations in the transfer characteristics of the ADC from a straight line. This departure impacts the consistency of quantization levels, which is notably adverse in applications demanding remarkable precision.

Quantization Errors: Quantization errors denote the distinction between the actual analog input and the nearest digital representation. Inherent to the process of analog-to-digital conversion, this phenomenon impacts the ADC's resolution.

Noise: Various forms of noise, encompassing thermal noise and quantization noise, introduce stochastic fluctuations into the signal. This interference can impede the ADC's capacity to distinguish minor signal variations, posing significant challenges, particularly within applications involving low-amplitude signals.

Jitter: Jitter refers to fluctuations in the timing of the sampling clock. Its primary impact is on high-frequency signals, introducing uncertainty in the temporal domain. Consequently, this uncertainty can translate into errors in the magnitude of the digitized signal.



Graph 8 Real distance and measured distance

Assignment 4

Experiment details

We used the digital-analog (DA) conversion function to output rectangular waves, rectangular waves, and sine waves from the DA conversion board. We completed the C++ program and observed the results using an oscilloscope.

Using the following variable names for the program code completion:

`nCount` for the number of iterations

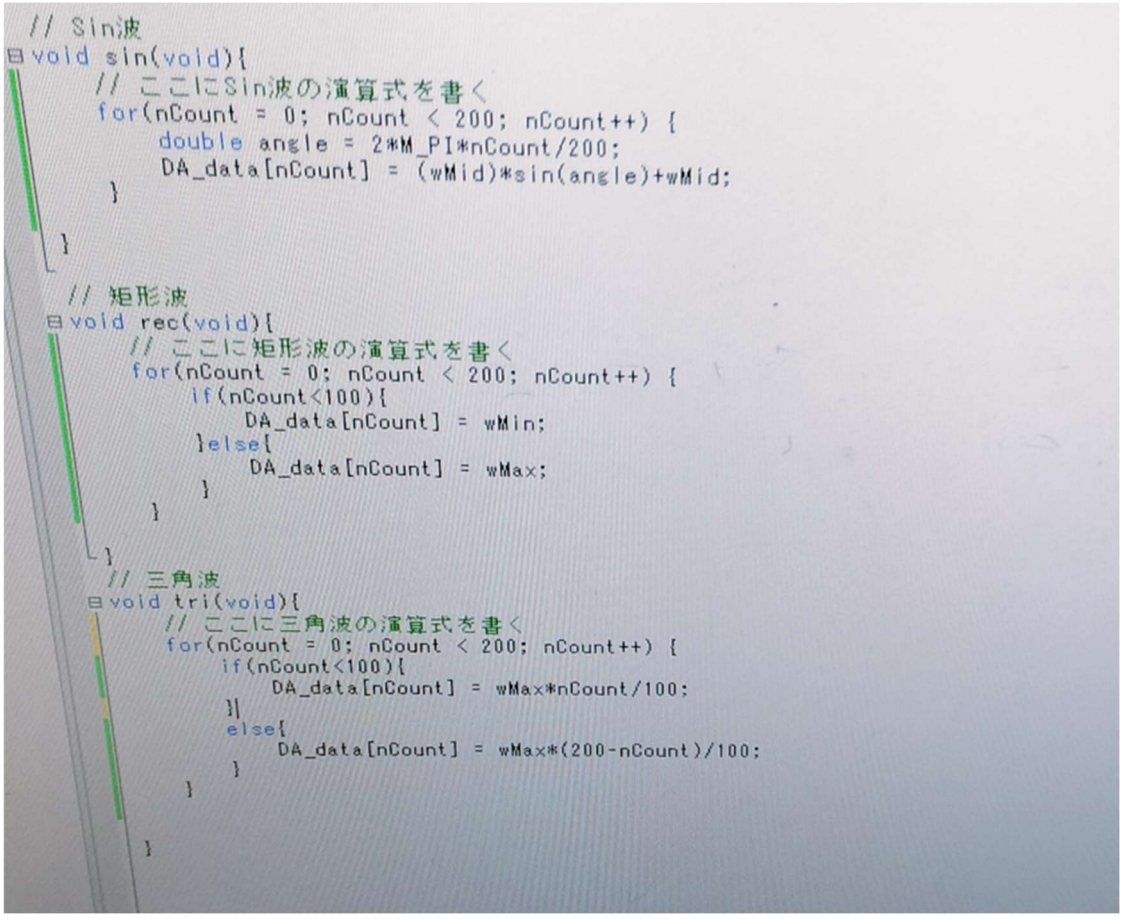
- `wMax` for the maximum output value (12-bit)

- `wMin` for the minimum output value

- `wMid` for the midpoint value

- `DA_data` for the output data

Program



```
// Sin波
void sin(void){
    // ここにSin波の演算式を書く
    for(nCount = 0; nCount < 200; nCount++) {
        double angle = 2*M_PI*nCount/200;
        DA_data[nCount] = (wMid)*sin(angle)+wMid;
    }
}
// 矩形波
void rec(void){
    // ここに矩形波の演算式を書く
    for(nCount = 0; nCount < 200; nCount++) {
        if(nCount<100){
            DA_data[nCount] = wMin;
        }else{
            DA_data[nCount] = wMax;
        }
    }
}
// 三角波
void tri(void){
    // ここに三角波の演算式を書く
    for(nCount = 0; nCount < 200; nCount++) {
        if(nCount<100){
            DA_data[nCount] = wMax*nCount/100;
        }
        else{
            DA_data[nCount] = wMax*(200-nCount)/100;
        }
    }
}
```

a) Program for the sine wave:
   In the sine wave function, we generated the sine wave asin(bx)+c, where the program sets the amplitude (a =) to wMid = (wMax+wMid)/2 where wMax = 4800 mV and wMin = 0; sets the angular frequency (b = 2π/T = (2)(3.14)/200); sets the vertical shift (c = wMid = 2400) because the calculations involve values only from 0V to 5V. The for-loop for 200 changes the value of nCount from 0 to 200 which generates a value for x = nCount in the sine function. The for-loop repeats the value generation for one period, generating a sine wave.
b) Program for the rectangular wave:
   In the rectangle wave function, a similar for-loop is executed. The if-statements generate a positive constant amplitude of wMax = 3000mV for half a period and an amplitude of wMin = 0 for the second half for x = nCount = 0 to 200.
c) Program for the triangular wave:
   In the triangle wave function, another if-statement is used to generate a function. The if-statements generate an amplitude of wMax*nCount /100 where wMax=5600 for the first half of a period and an amplitude of wMax*(200-nCount)/100 where wMax=5600 mV for the second hald of a period. So the slope of the triangular wave is wMax/100=5600/100 =56

Assignment 5

Sampling

Sampling is the process in which continuous-time signals or analog signals are converted into discrete-time signals (digital signals). The samples of the analog signals are taken at discrete time instants to obtain discrete-time signals. Sampling is associated with a term called sampling rate, which gives the number of samples per second, or sampling frequency.

Sampling frequency

Sampling rate or sampling frequency defines the number of samples per second (or per other unit) taken from a continuous signal to make a discrete or digital signal. [5] The sampling frequency is 1/ Δ t [s] or [Hz]. The smaller the value ofΔ t [s], the larger the amount of information that is sampled, is the which is preferred. However, if Δt is too small, it is meaningless.

The sampling frequency is significant in converting the analog signal to a digital signal without losing any special information. An adequate sampling frequency should be selected for recovering the spectrum of the analog signal from the spectrum of the discrete-time signal.

Sampling theorem (Nyquist theorem)

Sampling theorem or Nyquist theorem governs the sampling process for recovering the analog signal from the digital signal without loss of information. According to the sampling theorem, the sampling rate F_s is selected such that it is greater than or equal to $2f_m$samples per second, where f_m is the highest frequency of the continuous time or analog signal.

For analog signals to be used in computing, they must be converted to digital ones. However, digital electronics works in discrete numbers. To convert an analog signal to a digital one, measurements must be sampled at a regular frequency. The sample rate must be at least twice its frequency. This approach is used in digital audio and video to reduce aliasing, or the production of a false frequency.

A sample rate that is too low won't accurately depict the original signal; it will be distorted or have aliasing when reproduced. A rate that is too high will use more storage and processing resources than necessary.

The Nyquist theorem is used to locate the point where the right amount of information is gathered.

Quantization

Quantization comes after sampling as a crucial step in converting continuous analog signals to digital signals. A continuous set of values (like voltage levels) is quantized into a discrete set of values. During the analog-to-digital conversion process, each sampled value is matched with the closest value among a limited number of discrete levels.

Consider the sampled signal's amplitude as a continuous range. This range is split into quantized fixed intervals, each of which corresponds to a distinct digital code or level. The quantity of these intervals, or quantization levels, is determined by the quantization resolution, expressed in bits. There are 2^n different levels to which the sampled values may be translated with a n-bit quantizer. [6]

Quantization error

Quantization error is the difference between the analog signal and the closest available digital value at each sampling instant from the A/D converter. Quantization error also introduces noise, called quantization noise, to the sample signal. The higher the resolution of the A/D converter, the lower the quantization error and the smaller the quantization noise. The relationship between resolution (in bits) and quantization noise for an ideal A/D converter can be expressed as Signal to Noise (S/N) = -20*log (1/2^n) where n is the resolution of the A/D converter in bits. S/N is the signal to noise and is expressed in dB. This relationship can also be approximated as S/N = 6*n. Typical S/N ratios for ideal A/D converters are 96dB for 16 bits, 72dB for 12 bits, and 48dB for 8 bits. [7]

Sources and reference:

1. Analog-To-Digital Converters: How Does An ADC Work? | Arrow.com
2. Sources of Errors in ADCs
3. What is analog-to-digital conversion (ADC)?
4. The Sequence of Analog-to-Digital Conversion | Advanced PCB Design Blog | Cadence
5. Sampling rate (audio) - Glossary - Federal Agencies Digitization Guidelines Initiative
6. Fundamental Concepts: Sampling, Quantization, and Encoding
7. What is quantization error and how does signal to noise relate to this? | Tektronix
8. 5 Common Mistakes in Analog to Digital Conversion - CMA
9. Op-amp Integrator, Operational Amplifier Integrator
10. Differentiator Amplifier - The Op-amp Differentiator
11. Differential Amplifier - The Voltage Subtractor
12. Summing Amplifier is an Op-amp Voltage Adder
13. Op-Amp Integrator | CircuitBread

Conclusion

In this chapter, we learned about how operational inverting amplifiers work and the construction of such circuit and circuits that can be used for calculation using the inverting amplifier principle, such as summing circuits, subtraction circuits, integrator circuits, and derivative (differentiator) circuit. We learned the characteristics of each circuit, did circuit analysis and determined the gain. We also learned about contact sensors, analog to digital conversion (ADC) using contact sensors attached to operational amplifier and the AD conversion board and writing C ++ program to perform the AD conversion . We determined the relationship between voltage output and distance in AD conversion. We also learned the errors of AD conversion (sampling error and quantization error). Lastly, we learned and observed DA conversion by completing the C++ program and connecting the DA conversion boar

Second week program(AD.cpp)

```
#include <windows. h>
#include "AD_func. h"
#include "DA_func. h"
ADDATA addata,
DADATA dadata,
WORD DA_data[256],
double AD_data, // AD converted data entered
double Volt, // voltage
double Length, //length
int nCount, //Number of repetitions
WORD wMax= 0xfff, //Maximum value
WORD wMin=0, //Minimum value
WORD wMid= (wMax + wMin) / 2, //Medium value
//Edit the following from here!
//Third week: Assignment 2: AD conversion
void kadai1 (void) {
//Write voltage calculation equation here**********************************
//Volt= (AD value: data function)
//volt=:
volt= (10:4095.0)*AD_data-5.0;
}
//Third week: Assignment 3: Digital rangefinder
void kadai2 (void) {
//Write length calculation equation here**********************************
//Length= (voltage Volt function)
if(Volt<0.58){
```

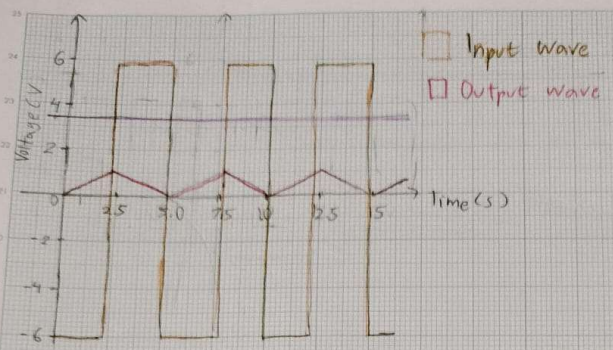
```
    length = -13.639+ 204.87 *(Volt) -793.61*(Volt*Volt)+1402.9*(Volt*Volt*Volt)-
914.21(Volt*Volt*Volt*Volt);
}
else{
length = 7.0649 + 2.8696*(Volt) -0.6987*(Volt*Volt) +0.0299*(Volt*Volt*Volt);
}
}
//Second week: Assignment 4 DA conversion
**********************************************
//saw tooth wave
void saw (void) {
     for (nCount = 0, nCount < 200, nCount++) {
     DA_data[nCount] = (WORD) nCount * wMax / 200,
      }
}
//Sine wave
void sin (void) {
//Write arithmetic expression for sine wave here
for(nCount=0; nCount<200; nCount++){
    double angle= 2*M*PI*nCount/200;
    DA_data[nCount] = (wMid)*sinAngle+wMid;
}
//square wave
void rec (void) {
//Write arithmetic expression for square wave here
for(nCount=0; nCount<200; nCount++){
  if(nCount<100){
     DA_data[nCount] =wMin;
```

```
  else{
      DA_data[nCount]=wMax;
}
}
//triangle wave
void tri (void) {
//Write arithmetic expression for triangle wave here
for(nCount=0; nCount<200; nCount++){
     if(nCount<100){
        DA_data(nCount)=wMax*nCount/100;
        }
      else{
      DA_data(nCount)=wMax*(200-nCount)/100;

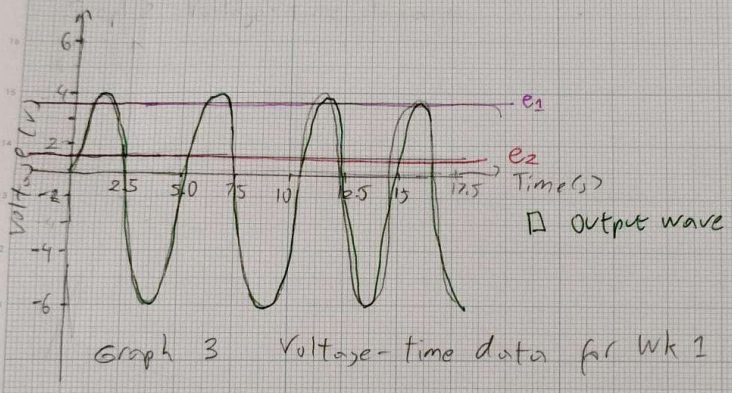
}



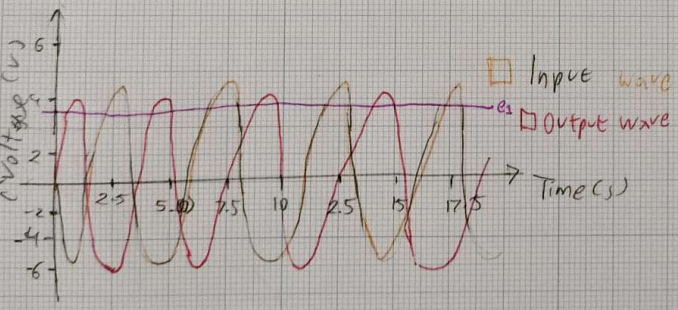
}
```



Graph 2 Voltage-time data for Week 1 Assignment 2



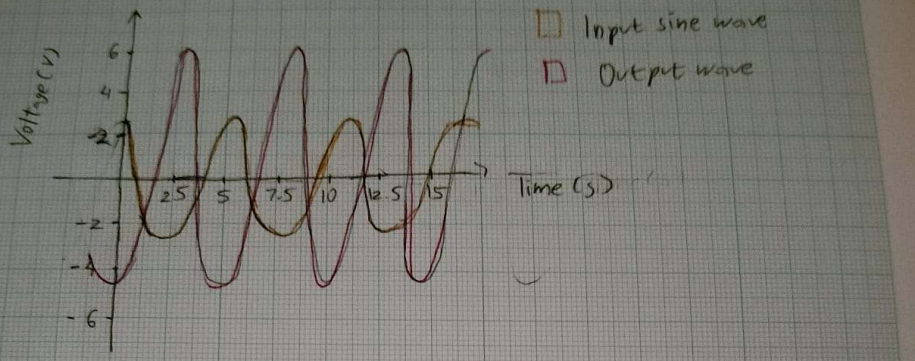
Graph 3 Voltage-time data for Wk 1 Assignment 3



Graph 4 Voltage-time data for Week 1 Assignment 4

Graph 5 Week 1 Assignment 5 Differentiator Circuit

(a) For input wave sine wave



(b) For input triangular wave

